Smart Library Request Workflow using ServiceNow

Capstone Project Documentation

Team No: 23BFA05428
Platform: ServiceNow
Domain: Academic Library Automation

1. Introduction

Project Title: Smart Library Request Workflow using ServiceNow

Academic libraries traditionally rely on manual registers, physical request slips, and in-person coordination between students and librarians to issue and track books. This approach is slow, error-prone, and offers little visibility into inventory status or borrowing patterns. The Smart Library Request Workflow addresses these challenges by digitizing the entire book request lifecycle on the ServiceNow platform, extending ServiceNow's workflow and automation capabilities beyond traditional IT service management into an academic operations use case.

The system introduces a structured, role-based process in which students raise book requests through a Service Catalog item, librarians review and approve or reject those requests, and the platform automatically updates inventory availability, notifies stakeholders, and records data for reporting. The result is a transparent, auditable, and efficient library management workflow suitable for colleges and universities of any size.

2. Project Overview

Purpose: To automate the end-to-end book request, approval, and inventory management process for an academic library using the ServiceNow platform.

Features: Book request submission via Service Catalog, role-based approval workflow, automated inventory and availability updates, request status tracking, email notifications, and borrowing trend dashboards.

The Smart Library Request Workflow is built around three primary user roles: the Student, who searches the catalog and raises requests; the Librarian, who reviews requests, manages inventory, and approves or rejects submissions; and the Administrator, who configures catalog items, workflow rules, and access controls. Students submit a request specifying the desired book title, author, and preferred pickup date. The request is routed to the Librarian queue for review.

Once a request is approved, ServiceNow's workflow engine automatically decrements the available copy count for that title, updates the request status to 'Approved', and triggers a notification to the student with pickup details. If no copies are available, the system can place the request in a waitlist state and notify the student once a copy is returned. Rejections, cancellations, and returns are all tracked as distinct workflow states so that the complete lifecycle of every book and request is auditable.

Beyond day-to-day transactions, the platform provides librarians and administrators with reporting and dashboard capabilities that surface borrowing trends, popular titles, overdue returns, and inventory health, enabling data-driven decisions about future acquisitions and resource allocation.

3. Architecture

Frontend: ServiceNow Service Portal / UI Pages

Backend: ServiceNow Business Rules, Script Includes, Client Scripts, Flow Designer

Database: ServiceNow Tables (Custom Book Catalog Table, Custom Request Table / RITM), storing book, request, and borrower data

The Smart Library Request Workflow follows a layered ServiceNow architecture that cleanly separates user interaction, business logic, and data persistence, ensuring the solution remains maintainable and extensible.

1. Presentation Layer (Frontend)

The frontend is delivered through the ServiceNow Service Portal, giving students a searchable book catalog and a request submission form, and giving librarians a dedicated workspace for managing pending requests and inventory. Client Scripts perform real-time validations, such as preventing a request from being submitted for a title with zero available copies, and dynamically show or hide fields based on user role.

2. Application Logic Layer (Backend Processing)

The backend layer implements all business rules and is composed of:

- Business Rules: Triggered on insert or update of request records, responsible for validating availability, updating copy counts, and setting request status.
- Script Includes: Reusable server-side JavaScript modules that encapsulate inventory lookup logic, due-date calculation, and fine/overdue evaluation.
- Flow Designer: Orchestrates the multi-step approval workflow, routing requests to the librarian queue, handling approvals/rejections, and triggering downstream notifications.

This layer guarantees that inventory counts, request states, and notifications remain consistent regardless of which interface (portal, mobile, or agent workspace) is used to interact with the system.

3. Data Layer (Database)

Library data is stored in ServiceNow Tables, primarily:

- A custom Book Catalog table (e.g., u_library_book) storing title, author, ISBN, category, and available copy count.
- A custom Book Request table (e.g., u_book_request) or the RITM table, storing requestor, requested title, request date, status, and approval details.
- Supporting tables for borrower history and return records.

4. Integration & Notification Layer

Once a request changes state (approved, rejected, ready for pickup, or overdue), the Notification Engine dispatches an automated email to the relevant student and, where applicable, the librarian, keeping all stakeholders informed without manual follow-up.

5. Reporting & Monitoring Layer

Dashboards and reports built on ServiceNow Reporting tools provide insights such as:

- Total requests raised and processed
- Approval versus rejection ratio
- Most frequently requested titles and categories
- Current inventory availability and low-stock alerts

4. Setup Instructions

Prerequisites: ServiceNow Developer Instance with admin access

Installation: Configure the Service Catalog item, create the book catalog and request tables, and configure business rules, client scripts, and the approval flow.

Step 1: Create Service Catalog Item

- Navigate to Service Catalog → Maintain Items
- Create a new catalog item named Library Book Request
- Add required variables such as: Book Title (Reference/Autocomplete), Author (Read-only), Preferred Pickup Date (Date Picker), and Remarks (Text Area)

Step 2: Create Backend Data Structure

- Navigate to System Definition → Tables
- Create a custom Book Catalog table (e.g., u_library_book) with fields: Title, Author, ISBN, Category, Total Copies, Available Copies
- Create a custom Book Request table (e.g., u_book_request) OR use the RITM table (sc_req_item) if extending the catalog flow, with fields: Requestor, Book Reference, Request Date, Status, Approved By

Step 3: Configure Client Scripts

- Navigate to Catalog Client Scripts
- Implement scripts for: availability check before submission, mandatory field enforcement, and dynamic display of author details once a book title is selected

Step 4: Configure Business Rules

- Navigate to System Definition → Business Rules
- Create rules to trigger after insert/update on the request table
- Implement logic for: decrementing available copy count on approval, incrementing it on return, and preventing duplicate active requests for the same student and title

Step 5: Implement Script Includes

- Create reusable Script Includes for backend logic such as: inventory availability checker, due-date and fine calculation engine, and utility methods for request state transitions

Step 6: Configure Approval Flow

- Navigate to Flow Designer
- Build a flow that routes new requests to the Librarian approval queue
- Configure branches for Approve, Reject, and Waitlist outcomes

Step 7: Configure Notifications

- Navigate to System Notification → Email → Notifications
- Create notifications triggered on request approval, rejection, and pickup readiness
- Ensure recipients are mapped to the requesting student and, where relevant, the librarian

Step 8: Configure Reporting and Dashboard

- Navigate to Reports → Create New Report
- Build dashboards showing total requests, approval trends, and inventory health
- Add widgets to the Service Portal for real-time visibility

Step 9: Testing the System

- Submit a test request via the Service Portal
- Verify record creation, availability decrement, notification delivery, and dashboard updates

5. Folder / Component Structure

1. Service Catalog Layer

- Contains the Library Book Request Catalog Item
- Defines user-facing form variables such as Book Title, Author, and Preferred Pickup Date
- Controls the user interaction flow in the Service Portal

2. Client-Side Logic

- Implemented using Client Scripts and UI Policies
- Responsible for real-time availability validation, dynamic field behavior, and basic client-side feedback before submission

3. Server-Side Business Logic

- Implemented using Business Rules, Script Includes, and Flow Designer
- Handles core backend operations such as inventory updates, request state transitions, and approval routing
- Ensures automation is executed consistently after record creation or update

4. Data Management Layer

- Uses ServiceNow Tables, either custom tables (u_library_book, u_book_request) or Service Catalog tables like sc_req_item
- Stores all transactional data including book metadata, request details, and borrower history

5. Reporting & Analytics Layer

- Built using ServiceNow Reports and Dashboards
- Provides insights such as total requests, approval ratios, and popular titles
- Supports operational monitoring and acquisition decisions

Key Structural Principle

The system follows a layered architecture model:

- UI Layer → Service Catalog + Portal
- Logic Layer → Client Scripts + Business Rules + Script Includes + Flow Designer
- Data Layer → ServiceNow Tables
- Insight Layer → Reports & Dashboards

6. Running the Application

Access Service Portal URL → Submit a book request via the catalog item → Backend approval and inventory processes run automatically via Business Rules and Flow Designer.

Step 1: Access the Service Portal

Users open the ServiceNow Service Portal using the instance URL and navigate to the Library Book Request catalog item available under the service catalog section.

Step 2: Fill the Book Request Form

The user enters the required details, including Book Title, Preferred Pickup Date, and Remarks.

- Mandatory fields are filled before submission
- Availability is checked in real time using Client Scripts
- Input data format is validated before submission

Step 3: Submit the Request

- A record is generated in the custom request table or RITM
- The request is assigned a unique request number automatically

Step 4: Backend Processing

- Business Rules execute automatically on insert/update
- The request is routed to the Librarian approval queue via Flow Designer
- Inventory availability is checked and reserved pending approval

Step 5: Librarian Review

The librarian reviews the request in their workspace and approves, rejects, or waitlists it. On approval, the available copy count is automatically decremented and the request status is updated.

Step 6: Notification Delivery

- An automated email notification is triggered once the request status changes
- The student receives confirmation with pickup details, or a rejection/waitlist message

Step 7: Monitoring and Verification

- Librarians and administrators track requests through the record view (request table or RITM)
- Reports and dashboards show request status and inventory analytics

7. API / Automation Logic

The Smart Library Request Workflow primarily uses ServiceNow's internal automation framework. Most of the business logic is implemented using Business Rules, Script Includes, and Flow Designer, ensuring that request processing, inventory management, and approval routing are handled natively within the platform.

1. Internal Automation (ServiceNow Platform)

- Triggering Business Rules on request creation and update
- Availability checking and inventory count adjustment based on request status
- Request number generation and record management
- Flow execution for the approval and rejection lifecycle
- Notification triggering after status changes

2. Overdue and Fine Calculation Logic

A Script Include compares the current date against the due date stored on an issued book record. If a book is not returned by its due date, the system flags the record as overdue and can compute an indicative fine amount based on a configurable per-day rate, which is then surfaced to the librarian for follow-up.

3. Automation Trigger Points

- Inventory adjustment runs after an approval Business Rule confirms sufficient copy availability
- Waitlist promotion runs when a returned-book Business Rule detects a copy becoming available
- This ensures inventory counts and request states never fall out of sync

4. Data Flow Summary

Book Request Record → Business Rule Trigger → Flow Designer Approval Routing → Script Include Inventory Check → Status Update → Notification Sent → Dashboard Refresh

8. Authentication

The Smart Library Request Workflow leverages ServiceNow's built-in Role-Based Access Control (RBAC) framework to ensure secure and appropriate access to library data and functionality.

1. Authentication Mechanism

User authentication is managed by the ServiceNow platform itself. Users must log in using valid ServiceNow credentials before accessing the Service Portal or backend modules. Session management and identity verification are handled by the platform's security layer.

2. Role-Based Access Control (RBAC)

Access is restricted using predefined and custom roles.

- Student Role (custom library_user): Can browse the catalog, submit requests, and view their own request history
- Librarian Role (custom library_admin): Can review, approve, or reject requests, and manage book inventory
- Administrator Role (admin): Full access to configuration, tables, scripts, and system settings

3. Data Security Controls

- Table-level security (ACLs): Restrict access to book and request records based on user role
- Field-level security: Sensitive fields, such as internal processing notes, are hidden from students
- Record-level security: Students can only view their own requests; librarians can view all requests
- Script protection: Server-side scripts ensure inventory logic cannot be bypassed from the frontend

4. Portal Security

- Service Portal access requires authentication
- Unauthorized users are redirected to the login page
- Form submissions are validated server-side to prevent unauthorized data manipulation

9. User Interface

The Smart Library Request Workflow provides a user-friendly interface built using the ServiceNow Service Portal, designed to simplify book discovery, request submission, and librarian review.

1. Book Catalog and Request Form (Service Portal)

- Searchable catalog listing available titles, authors, and categories
- A structured request form capturing Book Title, Preferred Pickup Date, and Remarks
- Dynamic validations using Client Scripts and UI Policies
- Real-time availability feedback before submission

2. Request Confirmation Page

After successful submission, users are redirected to a confirmation page displaying:

- Unique Request Number
- Requested book details and preferred pickup date
- Current request status

3. Librarian Workspace

Librarians access a dedicated list view of pending, approved, and rejected requests, along with quick actions to approve, reject, or waitlist a request, and to update inventory counts.

4. Dashboard Interface

An analytics dashboard implemented using ServiceNow Reporting tools provides:

- Total requests raised over time
- Approval versus rejection trends
- Most frequently requested titles and categories
- Current inventory levels and low-stock alerts

5. User Experience Design

- Simplicity: minimal steps required to raise a request
- Consistency: uniform layout across portal pages
- Responsiveness: compatible with different screen sizes
- Clarity: clear status labels and error messages

10. Testing

The Smart Library Request Workflow was validated using a structured manual testing approach to ensure correctness of the request flow, inventory logic, and notification delivery.

1. Test Scope

- Service Catalog Book Request Form
- Inventory Availability Logic
- Approval Flow (Business Rules and Flow Designer)

- Notification Delivery
- Dashboard Data Updates

2. Test Scenarios and Cases

Test Case	Input	Expected Output	Result
Request Submission	Valid book title and pickup date	Request record created successfully	Pass
Availability Validation	Book with zero available copies	System blocks submission / offers waitlist	Pass
Approval Processing	Librarian approves a valid request	Available copy count decremented automatically	Pass
Rejection Handling	Librarian rejects a request	Status updated and student notified	Pass
Return & Restock	Book marked as returned	Available copy count incremented; waitlist promoted	Pass
Notification Delivery	Successful approval/rejection	Email sent with request details and status	Pass

11. Known Issues

Despite successful implementation of the Smart Library Request Workflow, certain limitations exist due to platform constraints and process assumptions.

1. Manual Physical Handover

The system tracks requests and approvals digitally, but the physical handover of the book at the library counter is not automatically verified within ServiceNow.

2. Static Fine Configuration

Overdue fine calculation is based on a fixed per-day rate. Category-specific or membership-based fine variations are not currently implemented.

3. No Real-Time Payment Integration

The system does not include payment gateway integration for collecting overdue fines online; fine settlement is assumed to happen offline at the library desk.

4. Limited Scalability Testing

The system has not been tested under high-load conditions, such as large-scale concurrent requests during the start of an academic semester, so performance under heavy traffic is not fully validated.

5. No Barcode/RFID Integration

Physical book identification currently relies on manual title lookup rather than barcode or RFID scanning, which could otherwise speed up issue and return processing.

6. Single-Branch Assumption

The current data model assumes a single library branch. Multi-branch inventory and inter-branch transfer of books are not supported in this version.

12. Future Enhancements

The Smart Library Request Workflow can be extended further to improve usability, scalability, and real-world deployment readiness.

1. Barcode / RFID Scanning Integration

A scanning module at the library counter could automatically update issue and return records, reducing manual data entry and speeding up transactions.

2. Mobile App / Push Notifications

A companion mobile experience could deliver push notifications for approval status, due-date reminders, and waitlist availability.

3. Payment Gateway Integration

Integrating a payment gateway would allow students to pay overdue fines online, closing the loop on the borrowing lifecycle within the platform.

4. Multi-Branch and Inter-Library Support

Extending the data model to support multiple library branches and inter-branch book transfers would make the solution suitable for larger university systems.

5. AI-Based Recommendation Engine

A recommendation feature could suggest titles to students based on their borrowing history and academic course of study, improving engagement with library resources.